

cane). The sugar is then extracted through diffusion in hot water (beets) whereas cane juice is pressed out of the sugar-cane fiber using roller mills. In both instances the result is a solution or juice containing a percentage of sucrose, water, and impurities. The juice is then treated chemically with carbon and/or lime and filtered or decanted to remove most of the impurities. The resulting juice is evaporated and the sugar is crystallized; the result of this first crystallization is brown “raw” crystallized sugar that is covered in and saturated with its own molasses.

Separating the pure sucrose from the molasses involves a number of steps. The molasses is removed from the raw sugar by (1) spinning it at high velocity—a process called centrifuging; (2) redissolving it to separate more molasses; (3) recrystallizing; and (4) centrifuging again and rinsing. These steps are repeated until all the molasses is removed and more or less pure sucrose is attained.

Sugar Varieties

Granulated Sugar (Sucrose or Saccharose) – *Sucre Cristalisé*

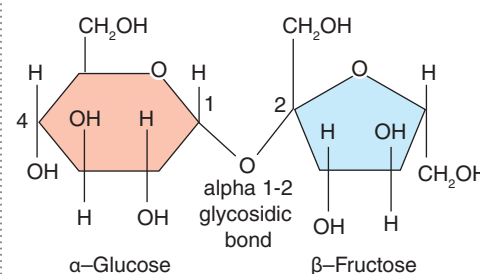
Whether derived from cane or beets, granulated sugar is more or less pure (99.8 percent) sucrose, which is a disaccharide, made of one glucose molecule and one fructose molecule. Granulated sugar, also known as “white,” “table,” “regular,” or simply “sugar,” is chosen for its crystal size which is large enough to crunch in the mouth but small enough that it dissolves readily in a solution (such as water if making a syrup or simply to sweeten a cup of coffee). Again due to its crystal size, when granulated sugar is used in pastry it delivers more friability (crunchiness) and flavor than powdered or superfine sugar; however, for the same reason, it produces dough that is a little harder to work with. Used in cakes, pastry, and confectionary, as well as in savory glazes, the range of its uses gives it the stature of an all-purpose sugar.

In pâtisserie, confectionary, and boulangerie alike, sugars of all types play an important role:

- In dough and batter, sugars weaken structure and improve tenderness by drawing moisture away from gluten and inhibiting its development; this ability to attract and retain water is known as sugar’s hygroscopic function.
- In addition to affecting texture—the friability or “crunchiness” of cookies, for instance—the hygroscopic function of sugar also acts as a preservative by denying bacteria the moisture it needs to grow within a given preparation.



Granulated sugar



Sucrose molecular formula

What's in a word?

Disaccharide

The word *disaccharide* is composed of the prefix *di*, which means two, and the root *saccharide*, which means sugar. This is clear enough considering disaccharides such as sucrose (table sugar), lactose (found in milk), and maltose (found in germinating grain) are each made up of two monosaccharides. In the case of sucrose, the bonded monosaccharides are fructose and glucose which themselves can be separated to make two different sugar products (see sucrose on p. 51 and glucose on p. 62). The accompanying diagram shows the fructose molecule, the glucose molecule, and the bond that makes the two form a sucrose molecule. Also included are the name and position of the atoms carbon, hydrogen, and oxygen that are the building blocks of the sucrose molecule.